

Every integer triple has a shortest Leech lift

Three disjoint Golay octads give a trace map $P : \Lambda \rightarrow \mathbb{Z}^3$. For every $t \in \mathbb{Z}^3$,

$$\min_{P\lambda=t} \|\lambda\|^2 = \frac{\|t\|^2}{4} + \mu(t \bmod 4).$$

Take $r_i \in \{0, 1, 2, 3\}$ and reduce 3 to -1 , giving $a(r)$. Then

$$\mu(r) = \begin{cases} 0 & r = (0, 0, 0), \\ 3 & r = (2, 2, 2), \\ 4 - \|a(r)\|^2/4 & \text{otherwise.} \end{cases}$$

In particular, the extra squared-norm cost is always at most $15/4$.

A finite seed proof, not a height extrapolation. Eight explicit norm-four seeds, code-preserving phase permutations and block sign changes give all 62 ordinary nonzero residues. A balanced dodecad realizes norm six at $(2, 2, 2)$, and parity proves that lower bound. Zero handles the remaining class.

Let $w_i = 21_{B_i}/\sqrt{8}$. Translation changes the i th trace by four and leaves the perpendicular component fixed. Hence

$$\lambda_*(t) = \nu_r + \sum_i \frac{t_i - a_i(r)}{4} w_i$$

is an explicit shortest inverse for every integer input.

The entire fibre is $\sigma(t) + \ker P$. Keeping that kernel preserves the marked state under subsequent maps; selecting a shortest representative does not remove the other states.

Full statement: An eight-seed proof of the shortest three-octad Leech trace inverse.